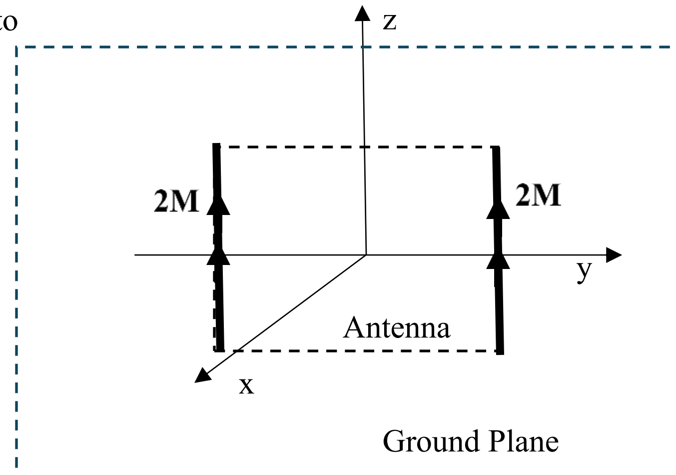
1. In class Notes 11 it was shown that a microstrip antenna radiates like two magnetic dipoles of length *a* spaced ½ wavelength apart with each having a constant current $2\mathbf{M}_s$. (a) Find the far electric field in the x-y plane for this two element magnetic dipole array antenna with two dipoles symmetric on the y-axis lying in the x = 0 plane as shown below. The coordinate system has been changed from that in notes 11 for mathematical convenience. (b) Make a sketch of the normalized electric field in the x-y plane.

Hint: Notice the two dipoles are parallel to each other and they must be considered as long linear antennas with a constant current. The magnetic currents are doubled in amplitude because the electric field has been doubled due to the ground plane and are assumed to be radiating into free space above the y-z plane.



2. (a) Find the fields for the TM_x mode in the partially filled waveguide in the notes. Use the simplified method shown in class for the TE_x mode, ie do not solve Helmholtz equation, just hypothesize the components of the wave function by applying boundary conditions. Number your equations and briefly explain what you are doing at each step as though this is a derivation. (b) Determine whether k_y or k_z or both can be zero and still have a propagating wave.

3. (a) Find the TE_z modes for the grounded dielectric waveguide mode cases given in the notes with $\mu_a = \mu_b = \mu_0$ and $\varepsilon_b = 2.4\varepsilon_0$, $\varepsilon_a = \varepsilon_0$, and, $t = \lambda_a/2$. Obtain the two equations necessary to find the propagation and attenuation constants. This problem will require careful construction of a figure in order to get accurate solutions to your transcendental equation.(b) Use graph paper to produce accurate tangent functions and a compass for constructing circles or generate the graph on a computer.
Note: When the slab is reduced in thickness the circles will reduce in radius reaching a point below which there is no intersection with the tangent lines and therefore no modes propagate. This is fortunate because a microstrip antenna above that particular thickness tends to create TE_z modes. If the dielectric is thin enough no modes are launched into the slab by the microstrip antenna and all energy leaving the antenna becomes a part of the antenna pattern. You can see from the example worked out in class that a TM_z mode will be produced down to zero slab thickness.